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代表了对一个人的智力表现的衡量，而较高的智商则意味着一个人的智力表现要优于大多数同龄人。

第四节经典测量理论（★★★）

考点1：经典测量理论的基本假设和真分数理论（熟悉）

经典测量理论也称真分数理论。它的基本思想是：任何测验分数都是由真分数和误差两部分组成，即：**X=T+E**。具体地讲，该理论认为，测量分数与其所代表的真分数之间是一种线性关系。

1. 在这个数学模型中各元素含义为

（1）观察分数：通过测验实际得到的分数，用X来表示。

（2）真分数：被测者在所测特质（如能力、知识、个性等）上的真实值，即无误差的测量分数，也就是从无数次测验中所得的平均数，用T表示，是一个理论上抽象的概念。

（3）误差分数：观察分数与真分数之差，即测量中的误差，用E来表示。

2. **CTT**的基本模型：观察分数与真分数是一种线性关系，可表示为**X=T+E**。

3. 关于CTT的三个基本假设，具体为：

（1）真分数具有不变性，即个体的某种特质具有稳定的特点。

（2）误差完全是随机误差（系统误差）。

（3）观察分数等于真分数与误差分数之和。

根据以上假设，观察分数的方差等于真分数的方差与误差分数的方差之和，可以由此推出信度、效度等多个重要的测量学概念和公式。

在测量学中，数学模型X=T+E是推导信度和效度公式的基础，它将复杂的测验分数内容用简单的形式表示出来，方便进行各种理论推导。

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第六章 项目分析与合成

第一节项目分析（★★★）

考点1：项目区分度的意义及计算方法（选择题、简答题、论述题、计算题）

项目区分度是指测验项目对被试心理特性的区分能力，常用P表示。P值越高，表示项目的区分能力越强，越能把不同水平的被试区分开来。在计算区分度时，首先需要确定效标，然后用如下两类方法计算：

1. 项目鉴别指数法（两端分组法）

（1）计算公式：根据测验总分区分出高分组和低分组，用公式**P＝R/N**分别求出两组在该题上的通过率。取**27%**为高、低**27%**组，然后计算两组通过率之差。

（2）两组被试在项目上的通过率之差即为项目鉴别指数，其数值越大，项目区分度越高。

2. 相关法：以项目分数与效标分数之间的相关作为项目区分度的指标。主要有以下几种：

（1）点二列相关：取值范围一般要在**0.5**。

（2）二列相关：适用于两列变量都是正态连续变量。

（3）ф相关：适用于两个变量都是二分变量的资料情况。

（4）多系列相关：适用于多分类变量。

（5）积差相关：适用于两个变量都是连续变量的情形。

（6）肯德尔和谐系数：等级数据。

相关法计算区分度时，既可以以测验总分做效标，也可以用外在效标分数，Z分数转换后，求相关系数。相关系数越高，表明项目越能区分出被试水平，项目区分度越好，因此可通过相关系数的大小来判断项目的区分能力，决定是否保留或删除该项目。

区分度与难度的关系：难度为中等时，区分度最大；难度越偏高或偏低，区分度越小。

第二节项目难度分析（★★★）

考点1：项目难度分析的含义（选择题、简答题）

项目难度是指被试完成项目作答任务时所遇到的困难程度，通常以通过率表示。项目难度与区分度密切相关，难度适中的项目区分度最高。难度越大，通过的人越少，区分度也越低。项目难度与区分度之间的关系可以用如下的关系式表示：**1**－**1**（D＞0 为正区分度，D＝0 为零区分度，D＜0 为负区分度），区分度越高，项目的质量就越好，项目的质量越好则区分能力越高。

区分度值的评价标准如下：

1. 项目区分度指数在一般人格测验中，教育测验中项目区分度的评价标准为：D 值在0.40 以上为非常优良。

2. 项目难度和区分度是相互联系的，项目难度对区分度影响较大，具体表现为：

（1）项目难度分布在**0～1**之间，越接近中间区分度越好。

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2. [illegible]WISC[illegible] 6—16 [illegible] WISC-V[illegible]WISC-V [illegible] 12 [illegible] 4 [illegible]

3. [illegible]WPPSI[illegible] 4—6 [illegible]WPPSI [illegible] 11 [illegible]

[illegible]

[illegible]★[illegible]

[illegible] 1[illegible]

[illegible]

1. [illegible]

2. [illegible]

[illegible]

（1[illegible]5.5[illegible]

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[illegible]

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1. [illegible]

2. [illegible]“[illegible]”[illegible]“[illegible]”[illegible]

3. [illegible]

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[illegible]

[illegible]★★★[illegible]

[illegible]1[illegible]16PF[illegible]MMPI[illegible]EPQ[illegible]

16PF[illegible]MMPI[illegible]EPQ[illegible]

1. [illegible]

（1）**16PF**[illegible]16[illegible]45[illegible]**16**[illegible]**10**[illegible]**16**[illegible]**16PF**[illegible]

（2）**MMPI**[illegible]**T**[illegible]**16**[illegible]**MMPI**[illegible]

（3）**EPQ**[illegible]**T**[illegible]**P**[illegible]**E**[illegible]**N**[illegible]**L**[illegible]**EPQ**[illegible]

2. [illegible]

3. [illegible]

[illegible]1[illegible]**16PF** [illegible] **EPQ** [illegible]16PF [illegible] **16** [illegible]EPQ [illegible] P[illegible]E[illegible]N [illegible] L [illegible]MMPI [illegible]

[illegible]2[illegible]**16PF** [illegible] **10** [illegible]**MMPI** [illegible] **EPQ** [illegible] **T** [illegible]

[illegible]3[illegible]**16PF** [illegible] **MMPI** [illegible] **16** [illegible]**EPQ** [illegible] 7[illegible]15 [illegible] 16 [illegible]

[illegible] 16PF [illegible] 16 [illegible]MMPI [illegible]EPQ [illegible] P[illegible]E[illegible]N [illegible] L [illegible]

[illegible] **2**[illegible]

[illegible] EPPS[illegible]16PF [illegible] MMPI[illegible]

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1. [illegible]

[illegible]1[illegible]

[illegible]2[illegible]“[illegible]”[illegible]“[illegible]”[illegible]

2. [illegible]

[illegible]1[illegible]

[illegible]2[illegible]

[illegible]

[illegible]★★★[illegible]

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1. [illegible]

2. [illegible]

3. [illegible]

4. [illegible]

5. [illegible]

6. [illegible]

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1. [illegible]

[illegible]1[illegible]**RIT**[illegible]10[illegible]5[illegible]3[illegible]2[illegible]

[illegible]2[illegible]**TAT**[illegible]

2. [illegible]

[illegible]1[illegible]**IAT**[illegible]

[illegible]2[illegible]

[illegible]3[illegible]

[illegible]

[illegible]★[illegible]

[illegible]**1**[illegible]

[illegible]

1. [illegible]

[illegible]1[illegible]EPPS[illegible]16PF[illegible]MMPI[illegible]

[illegible]2[illegible]

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1. [illegible]**MAS**[illegible] **MMPI** [illegible]

2. [illegible]**—**[illegible]**STAI**[illegible]**—**[illegible]

3. [illegible]**TAI**[illegible]**—**[illegible] **W** [illegible] **E** [illegible]W [illegible]E [illegible]

4. [illegible]**BAI**[illegible]

5. [illegible]**HAMA**[illegible]

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2. [illegible]

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（2）D [illegible]G[illegible]

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4. [illegible]

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（2[illegible]G[illegible]D[illegible]

（3[illegible]“[illegible]”[illegible]

[illegible]

[illegible]“G[illegible]—D[illegible]”[illegible]

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2. [illegible]

3. [illegible]

4. **Q** [illegible] 1 [illegible] 0 [illegible] Q [illegible] Q [illegible] **K** [illegible] **q** [illegible] **Q** [illegible]

5. [illegible]

6. [illegible]

7. [illegible]

[illegible] Q [illegible]

[illegible] **2** [illegible]

[illegible]

1. [illegible]

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[illegible]2[illegible]

[illegible]3[illegible]

[illegible]4[illegible]

[illegible]5[illegible]

[illegible]6[illegible]

[illegible]7[illegible]

[illegible]8[illegible]

2. [illegible]

[illegible]1[illegible]

[illegible]2[illegible]**DINA** [illegible]“[illegible]”[illegible]“[illegible]”[illegible]

[illegible]

[illegible]“[illegible]—[illegible]—[illegible]—[illegible]”[illegible]